Vrushank Saravade

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Education

Savitribai Phule Pune University — Pune, Maharashtra, India
Bachelor of Engineering in Information Technology, CGPA: 8.5/10 — *Aug. 2023 – May 2027*

- Currently in Fourth Year
- Relevant Coursework: Data Structures, Algorithms, Object-Oriented Programming, Database Management Systems

Experience

Open Source Contributor

prove-it-done-gemini | *Python, pytest*

- Developed a Python-based skill synchronization checker to identify missing or inconsistent files against an upstream repository
- Added automated test coverage with **6 passing test cases**

hiero-sdk-js | *JavaScript*

- Improved consistency and accuracy across **11 JavaScript SDK example files**
- Contributed **17 corrections** through a pull request merged into the official repository

Projects

Smart Job Matcher | *Python, FastAPI, React, Scikit-learn, NLTK* — GitHub | Live Demo

- Built an explainable **NLP** job matcher that scores resumes against job descriptions using **TF-IDF** and **cosine similarity** combined with skill-gap analysis
- Developed a skill-matching engine that reports matched and missing skills alongside a weighted match score
- Designed a modular **FastAPI** backend (routes, services, schemas) with a **React** frontend, deployed on Render

DevMatch v2 – Developer Networking Backend | *Python, FastAPI, MongoDB, JWT* — GitHub

- Built a **RESTful** backend API for a developer networking platform using **FastAPI**, with **MongoDB** (PyMongo) for persistent user and profile data
- Implemented **JWT**-based authentication with **bcrypt** password hashing and secured endpoints with authorization checks
- Designed a modular architecture separating routes, schemas, services, and database layers

AI Agricultural Decision Support System | *Python, Scikit-learn, CatBoost, LIME, React* — *In Progress*

- Designing an integrated decision-support platform combining crop recommendation, yield prediction, and price forecasting into a single workflow
- Building a **Multinomial Logistic Regression** model for crop recommendation using soil and environmental features with Five-Fold Cross Validation
- Developing **ensemble regression** models (Gradient Boosting, CatBoost, Bagging) to estimate crop yield from climatic and soil data
- Implementing **ARIMA**-based time-series forecasting for agricultural commodity price prediction
- Integrating Explainable AI using **LIME** to surface influential features behind individual model predictions

Publications

AI-Based Integrated Agricultural Decision Support System for Crop Recommendation, Yield Prediction and Price Forecasting Using Explainable Artificial Intelligence
Vrushank Saravade (Primary Author) | Accepted at International Conference on Intelligent Computing and Smart Systems (ICICSS 2026), Pune – to appear in conference proceedings

Technical Skills

Languages: Python, Java, JavaScript, HTML/CSS, SQL
Frameworks: FastAPI, Django
Databases: MongoDB
Tools: Git, GitHub, Postman, Uvicorn, VS Code
Concepts: REST APIs, Authentication & Authorization, Object-Oriented Programming, Data Structures & Algorithms, Agile SDLC, Debugging, Unit Testing, Database Fundamentals, AI/ML Fundamentals

Certifications

Python Full Stack Virtual Internship, EduSkills